

Siqi Gu

Nanjing, Jiangsu, China | +86 159 5191 1032
sikygu@163.com | [Google Scholar](#)



Research Interests

Intelligent software testing, LLM-based software engineering, and AI evaluation — with a focus on LLM-assisted unit test generation, search-based software testing (SBST), and automated testing for AI-based (multi-modal / text-to-image) software. My work has appeared in ACM TOSEM, ISSTA, and IEEE QRS.

Education

Nanjing University (2022.09 – 2026.09) — Ph.D. in Software Engineering, School of Intelligent Software and Engineering, supervised by Prof. Zhenyu Chen. Completed a one-year research exchange at University College London (UCL), working with Prof. Earl T. Barr on search-based software testing and LLM-based unit test generation (Feb 2025 – Feb 2026).

Nanjing University of Science and Technology (2019.09 – 2022.04) — M.S. in Computer Technology. Research focus: computer vision, multi-modal learning, and object detection.

Beijing University of Chemical Technology (2015.09 – 2019.06) — B.S. in Computer Science and Technology.

Publications

Peer-Reviewed Publications

1. **S. Gu**, Q. Zhang, K. Li, C. Fang, F. Tian, L. Zhu, J. Zhou, Z. Chen. “TestART: Improving LLM-based Unit Testing via Co-evolution of Automated Generation and Repair Iteration.” *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.
2. **S. Gu**, C. Fang, Q. Zhang, Z. Chen. “ACTesting: Automated Cross-Modal Testing Method of Text-to-Image Software.” *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 35(7):1–24, 2026.
3. Q. Zhang, Y. Shang, C. Fang, **S. Gu**, J. Zhou, Z. Chen. “TestBench: Evaluating Class-Level Test Case Generation Capability of Large Language Models.” *Journal of Systems and Software (JSS)* (arXiv:2409.17561).
4. Y. Shang, Q. Zhang, C. Fang, **S. Gu**, J. Zhou, Z. Chen. “A Large-Scale Empirical Study on Fine-Tuning Large Language Models for Unit Testing.” *Proceedings of the ACM on Software Engineering, Issue ISSTA*, 2(2):1678–1700, 2025.

5. **S. Gu**, Z. Lian. "A Unified RGB-T Crowd Counting Learning Framework." *Image and Vision Computing*, 131:104631, 2023.
6. **S. Gu**, Z. Liu, Q. Liang, Z. Deng, X. Xiao. "Test Case Reuse Method Based on Deep Semantic Matching." *2023 IEEE 23rd International Conference on Software Quality, Reliability and Security (QRS)*, 2023.
6. **S. Gu**, Z. Zong, F. Tian, Z. Chen. "DVTest: Deep Neural Network Visualization Testing Framework." *2023 10th International Conference on Dependable Systems and Their Applications (DSA)*, 2023.
7. **S. Gu**, J. Liu, Z. Hui, W. Liu, Z. Chen. "MetaA: Multi-Dimensional Evaluation of Testing Ability via Adversarial Examples in Deep Learning." *2022 IEEE 22nd International Conference on Software Quality, Reliability and Security (QRS)*, pp. 1004–1013, 2022.

Preprints, Under Review & In Preparation

- Q. Zhang, C. Fang, **S. Gu**, Y. Shang, Z. Chen, L. Xiao. "Large Language Models for Unit Testing: A Systematic Literature Review." *arXiv:2506.15227*, 2025.
- **S. Gu**, Q. Zhang, K. Li, C. Fang, E. T. Barr, Z. Chen. "No More Guesswork? When Does SBST Beat LLMs at Unit Test Generation." *IEEE Transactions on Software Engineering (TSE)*, under review.
- **S. Gu**, K. Li, Q. Zhang, C. He, Q. Cao, C. Fang, Z. Chen. "SUMMIT: Search-Based Unit Test Generation Improved by Large Language Model Integration." *Empirical Software Engineering (EMSE)*, major revision.
- **S. Gu**, C. Fang, Q. Zhang, F. Tian, J. Zhou, Z. Chen. "A Multi-Objective Unit Test Generation Method Based on Large Models and Code Pruning Techniques." *2026 International Conference on Quality, Reliability, and Security (QRS)*, 2026.
- Q. Zhang, Y. Shang, **S. Gu**, J. Zhou, C. Fang, Z. Chen, L. Xiao. "Multi-Agent LLM Collaboration for Unit Test Generation via Human-Testing-Inspired Workflows." *arXiv:2607.09101*, 2026.

Technical Skills

- **Programming:** Python, Java, C++
- **Research tools:** PyTorch, TensorFlow, Git, Linux
- **Research areas:** software testing, LLM evaluation, AI-based software engineering, search-based software testing (SBST)

Academic Service

- Reviewer, *IEEE Transactions on Software Engineering (TSE)*

Awards & Honors

- Expected Outstanding Graduate, Nanjing University, 2026
- National Scholarship for Doctoral Students, 2023
- Outstanding Graduate Student Model, Nanjing University, 2023